

## Lecture 22: Surface Integrals

### Background & Motivation

We've integrated functions along curves (line integrals) and over flat regions (double integrals). Today we integrate over curved surfaces in 3D. Surface integrals compute mass of a shell, and flux integrals measure how much a vector field flows through a surface. These are the ingredients we need for Stokes' and the Divergence Theorem.

### 1. Parametric Surfaces

#### Definition 1: Parametric Surface

A **parametric surface**  $S$  is the image of a continuous map  $\mathbf{r} : D \rightarrow \mathbb{R}^3$ , where  $D \subseteq \mathbb{R}^2$ . Each pair  $(u, v) \in D$  is sent to a point on  $S$  — think of latitude and longitude on a globe.

$$\mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle, \quad (u, v) \in D.$$

- **Tangent vectors:**  $\mathbf{r}_u = \frac{\partial \mathbf{r}}{\partial u}$ ,  $\mathbf{r}_v = \frac{\partial \mathbf{r}}{\partial v}$
- **Normal vector:**  $\mathbf{n} = \mathbf{r}_u \times \mathbf{r}_v$  (perpendicular to  $S$ ; nonzero where  $S$  is smooth)
- **Surface area element:**  $dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du dv$

#### Common Parameterizations

| Surface                                          | Parameterization $\mathbf{r}(u, v)$                                             | Domain                                    | $dS = \ \mathbf{r}_u \times \mathbf{r}_v\  du dv$ |
|--------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------|
| Graph $z = g(x, y)$                              | $\langle x, y, g(x, y) \rangle$                                                 | $(x, y) \in D$                            | $\sqrt{1 + g_x^2 + g_y^2} dA$                     |
| Sphere $x^2 + y^2 + z^2 = a^2$                   | $\langle a \sin \phi \cos \theta, a \sin \phi \sin \theta, a \cos \phi \rangle$ | $\theta \in [0, 2\pi], \phi \in [0, \pi]$ | $a^2 \sin \phi d\phi d\theta$                     |
| Cylinder $x^2 + y^2 = R^2$                       | $\langle R \cos \theta, R \sin \theta, z \rangle$                               | $\theta \in [0, 2\pi], z \in [c, d]$      | $R d\theta dz$                                    |
| Cone $z = \sqrt{x^2 + y^2}$                      | $\langle r \cos \theta, r \sin \theta, r \rangle$                               | $\theta \in [0, 2\pi], r \in [0, h]$      | $r\sqrt{2} dr d\theta$                            |
| Plane thru $P_0$ , dirs $\mathbf{a}, \mathbf{b}$ | $P_0 + s\mathbf{a} + t\mathbf{b}$                                               | $(s, t) \in D$                            | $\ \mathbf{a} \times \mathbf{b}\  ds dt$          |
| Rev. of $y = f(x)$ about $x$ -axis               | $\langle x, f(x) \cos \theta, f(x) \sin \theta \rangle$                         | $x \in [a, b], \theta \in [0, 2\pi]$      | $ f(x) \sqrt{1 + f'(x)^2} dx d\theta$             |

#### Special Case: $z = g(x, y)$

$$dS = \sqrt{1 + g_x^2 + g_y^2} dA$$

$$\mathbf{n} dS = \langle -g_x, -g_y, 1 \rangle dA \quad (\text{upward-pointing normal})$$

For downward normal, negate:  $\langle g_x, g_y, -1 \rangle dA$ .

## 2. Scalar Surface Integrals

### Definition 2: Surface Integral of a Scalar Function

The **surface integral**  $\iint_S f \, dS$  generalizes the double integral from a flat 2D region to a curved surface in  $\mathbb{R}^3$ . It sums values of  $f$  over  $S$  weighted by the true area element  $dS$ .

$$\iint_S f \, dS = \iint_D f(\mathbf{r}(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| \, du \, dv$$

For  $z = g(x, y)$ :

$$\iint_S f \, dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} \, dA$$

– **Applications:** mass ( $f = \rho$ ), average value on a surface

### Special Case: $f \equiv 1$ gives Surface Area

Setting  $f = 1$  in the surface integral turns it into the **surface area** of  $S$ :

$$\text{Area}(S) = \iint_S 1 \, dS = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\| \, du \, dv$$

For a graph  $z = g(x, y)$  over a region  $D$  in the  $xy$ -plane:

$$\text{Area}(S) = \iint_D \sqrt{1 + g_x^2 + g_y^2} \, dA$$

### Example 1: Mass of a Hemispherical Shell

Find the mass of the upper hemisphere  $x^2 + y^2 + z^2 = 9$ ,  $z \geq 0$ , with density  $\rho = z$ .

1. **Parameterize:**  $\mathbf{r}(\theta, \phi) = \langle 3 \sin \phi \cos \theta, 3 \sin \phi \sin \theta, 3 \cos \phi \rangle$

$$\theta \in [0, 2\pi], \phi \in [0, \pi/2]; \quad \|\mathbf{r}_\theta \times \mathbf{r}_\phi\| = 9 \sin \phi$$

2. **Set up:**  $\rho = z = 3 \cos \phi$

$$M = \int_0^{2\pi} \int_0^{\pi/2} (3 \cos \phi)(9 \sin \phi) \, d\phi \, d\theta$$

3. **Evaluate:**  $= 27 \int_0^{2\pi} d\theta \int_0^{\pi/2} \sin \phi \cos \phi \, d\phi = 27 \cdot 2\pi \cdot \frac{1}{2} = \boxed{27\pi}$

## 3. Flux Integrals

### Definition 3: Flux of a Vector Field Through a Surface

The **flux** of a vector field  $\mathbf{F}$  across an oriented surface  $S$  measures the net rate at which  $\mathbf{F}$  crosses  $S$  in the direction of the chosen unit normal  $\hat{\mathbf{n}}$ . If  $\mathbf{F}$  is a fluid velocity field, the flux is the volume

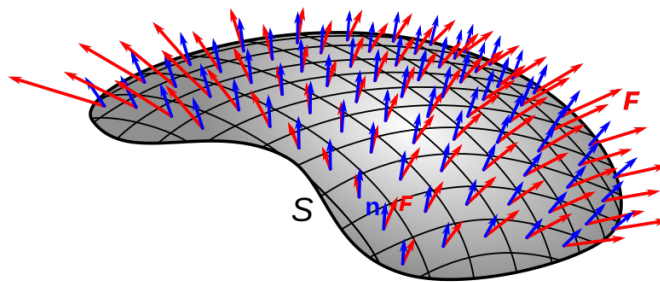
of fluid passing through  $S$  per unit time.

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) du dv$$

For  $z = g(x, y)$  with **upward** normal:

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D (-P g_x - Q g_y + R) dA$$

- **Interpretation: net flow of  $\mathbf{F}$  through the surface per unit time**
- Positive flux = net flow in direction of chosen normal
- **Orientation matters:** flipping normal negates the flux



Surface  $S$  with normal vectors  $\mathbf{n}$  (blue) and vector field  $\mathbf{F}$  (red). The flux integral measures how much  $\mathbf{F}$  flows through  $S$ .

### Example 2: Flux Through a Paraboloid

Find the flux of  $\mathbf{F} = \langle y, x, z \rangle$  through the paraboloid  $z = 1 - x^2 - y^2$ ,  $z \geq 0$ , with upward normal.

1. **Set up:**  $g(x, y) = 1 - x^2 - y^2$ , so  $g_x = -2x$ ,  $g_y = -2y$

$$P = y, Q = x, R = z = 1 - x^2 - y^2$$

Domain  $D$ :  $x^2 + y^2 \leq 1$  (disk of radius 1)

2. **Apply flux formula:**

$$\begin{aligned} \text{Flux} &= \iint_D (-y(-2x) - x(-2y) + (1 - x^2 - y^2)) dA \\ &= \iint_D (2xy + 2xy + 1 - x^2 - y^2) dA \\ &= \iint_D (4xy + 1 - x^2 - y^2) dA \end{aligned}$$

3. **Polar coordinates:**  $\int_0^{2\pi} \int_0^1 (4r^2 \cos \theta \sin \theta + 1 - r^2) r dr d\theta$

The  $4r^2 \cos \theta \sin \theta$  term integrates to 0 over  $[0, 2\pi]$ .

$$= \int_0^{2\pi} \int_0^1 (r - r^3) dr d\theta = 2\pi \left[ \frac{r^2}{2} - \frac{r^4}{4} \right]_0^1 = 2\pi \cdot \frac{1}{4} = \boxed{\frac{\pi}{2}}$$

### Example 3: Flux Through a Sphere

Find the outward flux of  $\mathbf{F} = \langle x, y, z \rangle$  through the sphere  $x^2 + y^2 + z^2 = a^2$ .

1. **Outward unit normal:** On the sphere,  $\hat{\mathbf{n}} = \frac{1}{a} \langle x, y, z \rangle$

2. **Dot product:**  $\mathbf{F} \cdot \hat{\mathbf{n}} = \frac{x^2 + y^2 + z^2}{a} = \frac{a^2}{a} = a$

3. **Integrate:**  $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S a \, dS = a \cdot 4\pi a^2 = \boxed{4\pi a^3}$

## Practice Problems

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**Problem 1.** Find the surface area of the plane  $z = 3 - x - y$  over the triangle  $0 \leq x \leq 1, 0 \leq y \leq 1 - x$ .

Answer:  $\frac{\sqrt{3}}{2}$

**Problem 2.** Find the flux of  $\mathbf{F} = \langle 0, 0, 1 \rangle$  upward through  $z = 1 - x^2 - y^2$  above  $z = 0$ .

Answer:  $\pi$

**Problem 3.** Compute  $\iint_S z \, dS$  where  $S$  is the cone  $z = \sqrt{x^2 + y^2}$ ,  $0 \leq z \leq 2$ .

(Hint:  $g_x = x/\sqrt{x^2 + y^2}$ ,  $g_y = y/\sqrt{x^2 + y^2}$ , so  $\sqrt{1 + g_x^2 + g_y^2} = \sqrt{2}$ .)

Answer:  $\sqrt{2} \int_0^{2\pi} \int_0^2 r \cdot r \, dr \, d\theta = \sqrt{2} \cdot 2\pi \cdot \frac{8}{3} = \frac{16\sqrt{2}\pi}{3}$

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**Problem 4.** Find the flux of  $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$  outward through the unit sphere  $x^2 + y^2 + z^2 = 1$ .

(Hint: on the unit sphere the outward unit normal is  $\hat{\mathbf{n}} = \langle x, y, z \rangle$ , so  $\mathbf{F} \cdot \hat{\mathbf{n}} = x^3 + y^3 + z^3$ . Use symmetry.)

Answer: 0 ( $\mathbf{F} \cdot \hat{\mathbf{n}} = x^3 + y^3 + z^3$ ; each term is odd in one variable on a symmetric sphere)

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**Problem 5.** Find the flux of  $\mathbf{F} = \langle z, x, y \rangle$  upward through the triangle cut from the plane  $x + y + z = 1$  in the first octant.

Answer:  $-Pg_x - Qg_y + R = -z(-1) - x(-1) + y = z + x + y = 1$ ; Flux =  $\iint_D 1 \, dA = \frac{1}{2}$